

CITY UNIVERSITY OF HONG KONG

Course code & title : EE3001 Foundations of Data Engineering

Session : Semester A 2026/27 Test 1

Time allowed : 75 minutes

This paper has 3 pages (including this cover page).

1. Do NOT turn over the question paper before the start of the test.
 2. Stop writing immediately when you are told to do so.
 3. The total marks is 100. Answer all questions.
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This is a closed-book test.

Students are allowed to use the following materials/aids:

Approved Calculators

Materials/aids other than those stated above are not permitted. Students will be subject to disciplinary action if any unauthorized materials or aids are found on them.

Each question below is 10 marks.

Qn 1

In Poker, “four of a kind” is a hand that contains four cards of one rank and one card of another rank. Derive the probability of drawing “four of a kind”.

Qn 2

A plane is missing and it is presumed to have gone down in five possible regions

R_1, R_2, R_3, R_4, R_5 with probabilities 0.1, 0.1, 0.2, 0.2, 0.4 respectively.

Let the probability that the plane will be found upon a search of the regions be 0.7, 0.7, 0.6, 0.5, 0.5 respectively.

What is the probability that the plane is in region 2 if a search has been made on region 1, 2, 3 and the plane has not been found.

Qn 3

Write down the definition of a) accuracy; b) precision. What are their meanings?

Qn 4

Simplify $Var(-X - 2Y + 3Z)$ as much as possible.

Qn 5

A random process (X, Y) for which X and Y are random variables records the following entries

(1, 2) (2, 1) (3, 6) (3, -6) (4, 4)

Derive the covariance between the two variables from first principle.

Qn 6

The mean and standard deviation of age at death of a country is 75 and 5 respectively. What is the upper bound of the probability of a person living to 115 years old?

Qn 7

You sample 30 people in the street of the country and it is found that the sample mean age is 45 years and the sample standard deviation is 5 years. Assume the sample standard deviation can approximate the true standard deviation. Find the bound in probability that the true mean is within 10 years from the sample mean.

Qn 8

Suppose you are equally likely to draw one of 5 coupons. If you draw 10 times, how many types of coupons are you expected to draw?

Qn 9

Derive the variance of a Bernoulli process with probability of success p from first principle. Remind you that a Bernoulli process is a process with two outcomes only, success or failure. A Bernoulli random variable may sometime appear as an indicator variable.

Qn 10

Your friend is very worried that she got a dreaded disease after testing positive in a medical test. Using your knowledge of Bayes' rule, what you can tell your friend that she can be less worried in certain cases. Explain clearly your reasoning process.

Appendix

$$P\{X \geq a\} \leq \frac{E[X]}{a}$$

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

$$P\left\{\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \geq \epsilon\right\} \leq \frac{\sigma^2}{n\epsilon^2}$$

$$P\{X - E[X] \geq \lambda\} \begin{cases} \leq \frac{\sigma^2}{\sigma^2 + \lambda^2} & \text{if } \lambda > 0 \\ \geq 1 - \frac{\sigma^2}{\sigma^2 + \lambda^2} & \text{if } \lambda < 0 \end{cases}$$

$$P\left[\left|\frac{1}{m} \sum_{i=1}^m \theta_i - \mu\right| > \epsilon\right] \leq 2\exp(-2m\epsilon^2/(b-a)^2)$$

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